

Alan Yu

AGENT INFRASTRUCTURE
ENGINEER

Internship · Paris · EN

CONTACT

✉ yualan077@gmail.com
🌐 github.com/Alan-Yu-2077
📍 Paris, France

PROFILE

I build the layer underneath the agent — turn loops, tool dispatch, sandboxing, and state that survives a restart. I don't train models; the LLM is a black box I call, and the engineering around that call is what I own.

CORE SKILLS

Runtime & scheduling State-machine turn loops · multi-agent async scheduling · three-state concurrency · checkpointed long-horizon state · full-turn tracing.

Execution & isolation Zod-first runtime validation at the tool boundary · timeouts and abort signals · capability sandboxing · SSRF-guarded fetch.

Production Strongly-typed full-stack TS · a shared wire contract that fails at compile time · fault tolerance · default-off flags for risky features.

Code — TypeScript · Python · C++ · Go

Runtime — Bun · Node · Electron · SQLite · WebSocket · MCP · Docker

LANGUAGES

Chinese (native) · English (fluent — working language)

Convention de stage available

Projects

agent-kernel — persona-free agent runtime · solo

TypeScript · [GitHub]

- **Turn engine:** a generic StateGraph loop (parse → build → stream → dispatch → append → finalize) driving streamed turns with interleaved tool use, bounded at eight tool iterations.
- **Tool kernel:** Zod-first defineTool factory and typed registry; dispatcher with three-state concurrency (safe-parallel / session-serial / global-serial), per-tool timeouts, abort discipline and a unified Result<T>.

Luna — long-running desktop agent · solo

Python prototype → TypeScript rewrite · [GitHub] [Demo]

- **Durable state:** three memory layers over one SQLite file with hybrid recall and a relevance floor; an offline pass distills each day into durable facts and reusable skills, so a restart loses nothing.
- **Isolation:** capability sandbox with canonicalized paths and an evaluator firewall — the agent may read the tests and configs that gate it, never write them; SSRF-guarded fetch, origin-checked transport.
- **Contract:** one Zod-typed WebSocket protocol shared by server and client — an unmirrored wire change is a compile error. 215 releases in seven weeks, each logged.

holdem-arena — multi-agent scheduling · solo

TypeScript

- N agents act at one table under contention — turn order, per-seat hidden state and two-channel output are the scheduler's problem. Built on agent-kernel: a second product off the same core.

Open Source

moeru-ai/airi — 46k ★ · 2 PRs merged

2026.06 – present

- Lifted Luna's web layer out and reshaped it for airi's stage-ui (#2066); fixed a background-process leak in their computer-use MCP (#2006). Two more in review.

moeru-ai/velin — 1 PR merged

2026.07

- Exposed a prompt component as an LLM tool (#49).

Experience

Campus Partner, Lead · ByteDance

2025

- Ran hackathons and campus vibe-coding workshops at Chang'an University. Named **Outstanding Campus Partner of the Year**.

Technical Intern · E-class, Chang'an University

2024.10 – 2025.02

- Selected **1 of 12**. Owned data collection, cleaning and ingestion for a campus RAG assistant — browser automation with pagination and resume-on-restart.
- Picked up how an agent is actually structured from the engineers around me. This is where everything above started.

Education

BSc Computer Science · EPITA, Paris

2026.09 – 2027.06 (expected)

Chang'an University × EPITA · Sino-French 3+1 joint programme

BEng Artificial Intelligence · Chang'an University

2023.09 – 2026.07